11 March 1977
SUPERSEDING
MIL-P-21904(WP)
21 January 1959
(see 6.3)

MILITARY SPECIFICATION

PARACHUTES FOR UNDERWATER MINES; GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general requirements for the production of parachutes for use with underwater mines.

1.2 Classification. Parachutes covered by this specification shall be of the following types (see 6.2 and 6.3):

Type III - Ring Slot
Type IV - Cross
Type V - Ribbon

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, Md. 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1350

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MILITARY

MIL-P-116	Preservation-packaging, methods of
MIL-B-131	Barrier Materials, Water Vapor-proof, Flexible, Heat-sealable

STANDARDS

FEDERAL

FED-STD-4	Glossary of Fabric Imperfections

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-849	Inspection Requirements, Definitions and Classification of Defects for Parachutes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General

3.1.1 Construction and materials. Parachutes shall be manufactured in accordance with applicable Naval Sea Systems Command Lists of Drawings and all documents listed thereon.

3.1.2 Overage textile materials. Unless otherwise allowable in an applicable textile specification, all textile materials two years or older from date of yarn manufacture shall not be used without the approval of the procuring agency.

3.1.3 First article sample. When specified (see 6.2), the contractor shall furnish a first article sample for testing and approval. This sample shall be manufactured in accordance with 3.1.1 using the fabrication procedures and facilities proposed and used for production. Acceptance of the sample qualifies the fabrication procedures and the specific facilities by which manufacture and test of the sample is accomplished. After manufacture and acceptance of the sample, no changes shall be made in the fabrication procedures without the approval of the procuring agency. When new or altered facilities are used in production, a new sample shall be submitted at the discretion of the procuring agency for acceptance as specified in 4.3.

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3.2 Layouts.

3.2.1 Working surfaces. All bolts or rolls of cloth, ribbons, tapes, lines, and webbing shall be unrolled on to or into clean smooth flat working surfaces or tables large enough to accomodate their subsequent relaxation, marking, and cutting operations.

3.2.2 Relaxation. Prior to any marking or cutting of the unrolled items of 3.2.1, the material shall be allowed to relax for a minimum of 24 hours to its true unstressed dimensions.

3.3 Markings.

3.3.1 Marking materials. Unless otherwise specified on applicable drawings, materials used for marking and indexing shall not be deleterious to the parachute's material or the serviceability of the parachute.

3.3.2 Marking of cloth and ribbons. Cloth and ribbon materials laid out in accordance with 3.2 shall be marked and indexed to the specified dimensions while in the relaxed condition.

3.3.3 Marking of lines, tapes and webbing. Line, tape and webbing materials laid out in accordance with 3.2 shall be marked and indexed to the specified dimensions while under tension as specified on applicable drawings.

3.4 Cutting.

3.4.1 Single whole pieces. Gore tapes and ribbons shall be cut from and as single whole pieces.

3.4.2 Textile defects. Gore panels, tapes and ribbons shall be cut so as not to include:

(a) Any defects which would be detrimental to the parachute's strength, durability and use.
(b) Any of the defects which are allowable in the material specifications, such as: broken picks, broken warp ends, floats (in either the warp or filling), holes, mispicks, smashes, spots of oil, thin places (due to starting or stopping of the loom or to pickouts), torn selvages, wavy selvages, etc.

3.5 Stitching. Stitching shall conform to the size and type specified on applicable drawings and the following requirements.

3.5.1 Stitching pattern. All stitching shall be evenly spaced, unbroken, and sufficiently tight so as to lie smoothly and evenly on the sewn surfaces. The stitching shall not:

(a) Loop or project above the sewn surface.
(b) Cause puckering or gathering of the cloth.

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3.5.1.1 Stitch repairs. Defective stitching, as described in 3.5.1, shall be repaired as follows:

(a) Broken, skipped, split and loose stitches shall be repaired by resewing adjacent to the faulty stitches.
(b) Tight stitches shall be removed and the seam resewn.
(c) Unless otherwise specified (see 3.5.2 (b)), all repairs shall extend for two inches beyond each end of the faulty segment.

3.5.2 Seams and hems. Unless otherwise specified on applicable drawings, seam and hem stitches shall not be within 1/16 inch of the edges of the seam. Defective stitching shall be repaired as follows:

(a) If the total length of repairs does not exceed one-half of the seam length, the defect may be repaired by a single resewing of the defect so as to include the defect length plus 2-inch overlaps at both ends.
(b) If the total length of repairs exceeds one-half of the seam length, the defect shall be repaired by removal of the stitches and resewing the seam.
(c) All repaired seams shall conform to 3.5.1.

3.5.3 Seam integrity. At all points along the main seams, each row of stitches shall pass through all layers of the material.

3.6 Trimming. Trimming, gathering and sewing of the raw edges of material at the seams shall be in accordance with the applicable drawings.

3.7 Assembly of suspension lines, tapes and webbing. All suspension lines, tapes and webbing shall be assembled so that all markings are within ± 1/4 inch of their indicated positions on the canopy.

3.8 Workmanship. The completed parachute shall be free from dirt, wax, grease and other foreign matter. Necessary precautions shall be taken during manufacture to ensure these conditions. The fact that repairs will result in an acceptable item does not mean that the use of repairs shall not be controlled. Corrective actions shall be taken to eliminate or control repairs and the maintenance of quality as prescribed by this specification and the parachute's drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article sample is required (see 6.2 and 6.4), ten (10) parachutes shall be manufactured and submitted to an activity designated by the procuring agency. This sample shall be inspected by the contractor and the designated activity for conformance to all of the requirements specified in this specification and applicable drawings (see 3.1.3). Further production of the parachutes by the contractor prior to approval of the procuring agency shall be at the contractor's risk.

4.4 Quality conformance inspection. Unless otherwise specified herein and on applicable drawings, the parachutes shall be 100% inspected. The quality conformance inspection shall consist of the inspections of 4.5.1 and 4.5.2. Failure to meet the applicable qualification requirements of 4.5 will result in rejection of the parachute.

4.5 Inspections.

4.5.1 Inspections to drawing requirements. The parachute, its materials, and its components shall be inspected to its drawing requirements in accordance with one of the following procedures, as applicable:

(a) When Classification of Characteristics (CC's) are specified in the parachute's drawing package, the inspection shall be in accordance with the CC's.
(b) When CC's are not specified in the parachute's drawing package, the inspection shall be in accordance with the Classification of Defects specified in the applicable table of MIL-STD-849 modified to include the dimensions and tolerances specified in the drawing package.

4.5.2 Inspections to specification requirements. The parachute shall be visually inspected to the requirements of Section 3, and defective items disposed of as given in Table 1.

5. PACKAGING

5.1 Preservation - packaging.

5.1.1 Level A.

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Table 1. Disposition requirements for inspection failures.

Defect	Procedures and Disposal
Fabric 2 years or older.	Examination of records; reject fabric.
Deleterious marking material.	Certificate of Compliance; reject material and unserviceable parachute.
Gore tapes and ribbons not single whole pieces.	Reject defective parachute.
Fabric defects present.	Reject defective parachute.
Stitching loose.	Reject defective parachute.
Stitching tight.	Reject defective parachute.
Stitching repairs improper.	Reject defective parachute.
Stitching within 1/16 inch of seam edge.	Examine on illuminated glass top table; reject defective parachute.
Stitch repairs exceed 1/2 length of seam.	Reject defective parachute.
Stitching does not pass through all layers.	Reject defective parachute.
Assembly of suspension lines, tapes and webbing are not within ± 1/4 inch of indicated positions.	Reject defective parachute.
Parachute is not free from dirt, wax, grease or foreign matter.	Reject defective parachute.
Parachute is extensively repaired.	Reject defective parachute.

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5.1.1.2 Unit packaging. Each parachute shall be individually packaged Method IA-15 (less preservative application) conforming to the requirements of MIL-P-116.

5.1.1.2.1 Unit container. The unit container shall be fabricated in accordance with the requirements of PPP-B-636, Style optional, Class domestic, variety SW, 275 lbs. minimum bursting strength. The individually packaged parachute shall be placed in a unit container which shall be inserted into a water-vaporproof bag, fabricated from Class 1 barrier material conforming to the requirements of MIL-B-131, and closure heat sealed. (Prior to inserting unit container into the water-vaporproof bag, all corners of the container shall be blunted).

5.1.2 Level B. (Not applicable)

5.1.3 Level C.

5.1.3.1 Unit packaging. Each parachute shall be individually packaged Method III, in accordance with the requirements of MIL-P-116.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior containers. Parachutes, prepared as specified above for Level A protection, shall be packed in multiples, in a Class weather-resistant, variety SW, Style optional, fiberboard box conforming to the requirements of PPP-B-636, W5c compliance symbol.

5.2.1.2 Overpacking. Each fiberboard carton, containing a multiple of packaged parachutes, shall be overpacked in an overseas type wood-cleated plywood box or in a type 2, 2-1/2, 3 or 4 nailed wood box conforming to the requirements of PPP-B-601 or PPP-B-621, respectively.

5.2.1.3 Lumber. When nailed wood boxes are used, lumber shall be of one (1) inch nominal thickness, surfaced on both sides, conforming to Groups I, II, or III woods of box specification. The gross weight in a given shipping container shall not exceed one-hundred (100) pounds.

5.2.2 Level B. (Not applicable)

5.2.3 Level C.

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5.2.3.1 Shipping Containers. Parachutes shall be packed in a manner which will provide protection from damage during direct shipment from the contractor or supply source to the first receiving activity for immediate use. The type of containers used shall conform to applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Normal markings. In addition to any special marking required by the contract or order, all container markings shall be in accordance with MIL-STD-129.

5.4 Government procurements. Unless otherwise specified in the contract or order, the parachutes covered by this specification shall be delivered in appropriate Parachute Packs as stipulated by the Naval Sea Systems Command. When there is direct purchase by the Government of subject parachutes, the above packaging requirements shall be required.

6. NOTES

6.1 Intended use. Parachutes produced in accordance with this specification are intended for use with underwater mines.

6.2 Ordering data. Procurement documents should specify the following:

a. - Title, number and date of this specification
b. - Type of parachute required (see 1.2)
c. - Applicable lists of drawings (see 3.1.1)
d. - Quantity of parachutes required
e. - Whether first article sample required (see 3.1.3 and 6.4)
f. - Name and address of activity to inspect the first article
g. - Level of packaging and packing required (see 5.4)

6.3 Correlation data. Types I and II parachutes of MIL-P-21904(WP) are obsolete and have been dropped from MIL-P-21904A(OS). Types IV and V designations have been added to this specification to preserve the continuity of designations and to prevent misinterpretations in parachute documentation, drawings and procurements.

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6.4 First article. Submission of the first article sample may be waived by the procuring activity (see 6.2 e) when the manufacturer is producing the parachutes for prior contracts. Production procedures and facilities must not be changed (as required in 3.1.3) when the first article sample is waived. When a first article sample is required it will be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulations. The first article should be a pre-production sample and should consist of ten units. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, tests, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - OS

Preparing activity:
Navy - OS

Project 1350-N033

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE MIL-P-21904A(OS)
PARACHUTES FOR UNDERWATER MINES: GENERAL SPECIFICATION FOR

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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